Apply to be a Science Corps Fellow - Entry #3406

Name

Keiko Wilkins

Email

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Gender

Female

Residence: City (State), Country

Honolulu, HI

Nationality

USA

When did you or do you expect to complete your PhD?

December 2025

Please tell us briefly about your university education (institution, city, country, year)

Miami University (Oxford, OH, USA January 2018) B.S. in Biological Sciences with a minor in Chinese
Miami University (Oxford, OH, USA January 2020) M.S. in Biology with an Applied Statistics Certificate
University of Hawaii at Manoa (Honolulu, HI, USA December 2025)PhD in Marine Biology

Why would you like to be a Science Corps Fellow?

As her tail glided across the tile floor, the light scratchy sound alerted me to her approach. Cautiously surveying her surroundings, she would stop every couple of feet before continuing on her original path. Every so often, you would hear her hiss if you caught her off guard. I would walk over to her, rub her rough, scaly back and wait to see what she would do. Most of the time, she would sit still and smile at me with her long, pointed teeth or she would bob her head slightly up and down, signaling that she was thirsty. My first pet was “Smiley”, a 4ft American Alligator.

I spent the first 18 years of my life standing behind my parents’ captive born reptile store counter selling and educating thousands of people from different countries and backgrounds on the proper husbandry of the animals I grew up taking care of and breeding. From adults purchasing their 100th python to children wanting their first quarter-sized turtle, I educated them all. My dad and I also presented educational programs on reptiles and amphibians discussing human impacts on their survival to children ages 3-18 at schools and summer camps. These early experiences shaped my identity as a science educator long before I pursued formal academic training.

This foundation led me to want to pursue a degree in science education in college. I pursued advanced biology degrees at Miami University and the University of Hawai’i at Mānoa instead as an alternative pathway to teach while also conducting research. While my academic training strengthened my scientific skills, my motivation has remained rooted in education and community engagement. Science Corps' emphasis on teaching, educational resource development and capacity building aligns with how I approach science, as a collaborative process grounded in place-based context.

I have had the opportunity to teach college-level biology laboratory courses as a teaching assistant and develop place-based educational resources through community engagement in the Pacific. However, I have consistently wished for more opportunities to focus on teaching during graduate school. I am drawn to the Science Corps Fellowship because it allows me to center education, learn from a new community, and contributing to locally relevant, community-based science learning. At this stage in my career, Science Corps represents a natural next step in my commitment to teaching science in ways that are accessible and meet community needs.

In addition to your studies, please describe any prior experience that may help you as a Science Corps Fellow.

Teaching and community engagement were central priorities throughout my PhD. Having grown up in the landlocked state of Ohio with limited exposure to marine science, I am deeply aware of how access and representation shape students' perceptions of who can pursue science. This motivated me to focus on education and outreach, using my experiences to show students that science careers are attainable and relevant to their lives.

While at the University of Hawai'i at Mānoa, I received the prestigious National Oceanic and Atmospheric Administration (NOAA) Dr. Nancy Foster Scholar, an experience that allowed me to work across Pacific Islands Region National Marine Sanctuaries while prioritizing classroom-based teaching and community engagement. In 2023, I visited over 30% of the elementary, middle and high schools in American Sāmoa, delivering interactive lessons on coral reef ecosystems and plastic pollution. I collaborated with educators and sanctuary staff to develop hands-on, experiential learning activities that encouraged students to ask questions, collect observations, and connect scientific concepts to their local environment.

In 2024, I secured a small grant to support the development of teacher resources and additional classroom visits in partnership with the American Sāmoa Department of Education and sanctuary staff. This work strengthened my skills in educational resource development, adapting lessons for different age groups, and integrating place-based learning into science education. These experiences also laid the groundwork for student-centered, community-based research activities, where students participated in data collection and inquiry relevant to their own communities.

In addition to my work in the Pacific Islands, I have conducted school visits throughout Ohio, visiting seven schools to share my pathway into marine science, introduce STEM and environmental science careers, and teach foundational concepts about coral reefs. I emphasized connections between inland communities and the ocean, helping students understand how ocean systems influence their daily lives, even far from the coast. Together, these experiences have prepared me to teach science in diverse cultural contexts, foster curiosity and confidence in students, and support community-based science projects as a Science Corps Fellow.

Where do you imagine yourself professionally in the future? Check all that apply.

Scientist in academia
In science outreach
Other

Please expand on your selection above and describe where you imagine yourself professionally in the near (~2 year) and long-term (~10 year) future.

My short-term goal is to advance my expertise in coral genomics while continuing to mentor and teach. I plan to begin a postdoctoral position in Summer or Fall 2027 that will allow me to continue independent research on corals while integrating hands-on mentoring for students. I thoroughly enjoy research, teaching, and mentorship, and I aim to continue integrating all three into my career.

During this period, I will remain actively engaged in mentoring underrepresented minority students in STEM. For the past four years, I have worked for the international non-profit Black Women in Ecology, Evolution, and Marine Science as the Community Coordinator. This organization builds community, provides mentorship, and supports professional development for Black women in environmental and marine sciences. In the short term, I aim to continue strengthening support for early-career scientists and broadening access to research training opportunities. Together, these efforts will allow me to further develop my skills as a researcher, educator, and mentor during this critical career stage.

My long-term goal is to secure a research position as a faculty member at a university or as a research scientist at NOAA or another governmental agency. In this role, I aim to lead research programs focused on microplastic impacts on marine life, with an emphasis on biologically and culturally important species. Given the global scale and long-term persistence of microplastic pollution, I am driven to produce research that informs conservation, management, and mitigation strategies. I intend to continue prioritizing both research and teaching throughout my career, ensuring that science benefits diverse communities. I believe that the Science Corps Fellowship will allow me to gain additional teaching experience, further developing my skills as a well-rounded scientist prepared to mentor and engage diverse communities.

Please upload your CV (.pdf)

WilkinsKeiko_CV_2025_teaching.pdf

How did you find out about Science Corps?

An email list in your institution

Science Corps